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**Operating Instructions
for
Triple Grating Turrets
SP-750**

Supplemental Instructions for Acton Models With Kinematically Mounted Triple Grating Turret

This Acton series monochromator/spectrograph is supplied with a kinematically mounted triple grating turret assembly that is interchangeable in the monochromator/spectrograph without re-alignment.

The grating turret is removed from the instrument for shipment. To install the turret, the following procedure is recommended:

CAUTION: THE OPTICAL SURFACES IN THE MONOCHROMATOR/SPECTROGRAPH ARE EXTREMELY DELICATE AND CAN BE PERMANENTLY DAMAGED BY CONTACT WITH ANYTHING. DO NOT TOUCH, TALK OR BREATHE ON THE OPTICAL SURFACES.

1. Locate the storage container with the grating turret to be installed. Become familiar with the components as labeled on Figures 1, 2 and 3 in this section.
2. Scan the instrument to grating #1 and the 0000 wavelength position.
3. Place the monochromator in a clean, dust free area.
4. Remove the screws from the monochromator access cover and remove the access cover. Become familiar with the grating turret and its components.
5. Remove the plastic cover from the grating turret storage container. Loosen the 3 grating turret clamping screws. **NOTE:** These screws are captive. **CAUTION:** Do not hit the grating surfaces as permanent damage will be caused.
6. Carefully grasp the grating turret by the top plates. **CAUTION:** Do not touch the grating surfaces. Lift the grating turret out of the storage container.
7. Locate the grating turret # on the grating turret assembly. Assure the monochromator is as grating #1 and 0000 wavelength position. Refer to Figure #1 and place the grating turret on the grating spindle with the turret # or red dot located as shown.
8. Assure the turret is properly seated, the ball in the cone hole and the rod in the vee groove.
9. Lightly tighten clamp screw #1, then #2 and #3. (**NOTE:** To assure maximum reproducibility, the clamping screws must be tightened in this sequence and to 3 inch pounds of torque.) Tighten screw #1, to 3 inch pounds of torque, then #2 and #3. The grating turret is now installed. Refer to the instruction manual for operation.

INTERCHANGING TURRETS

FIGURE 1

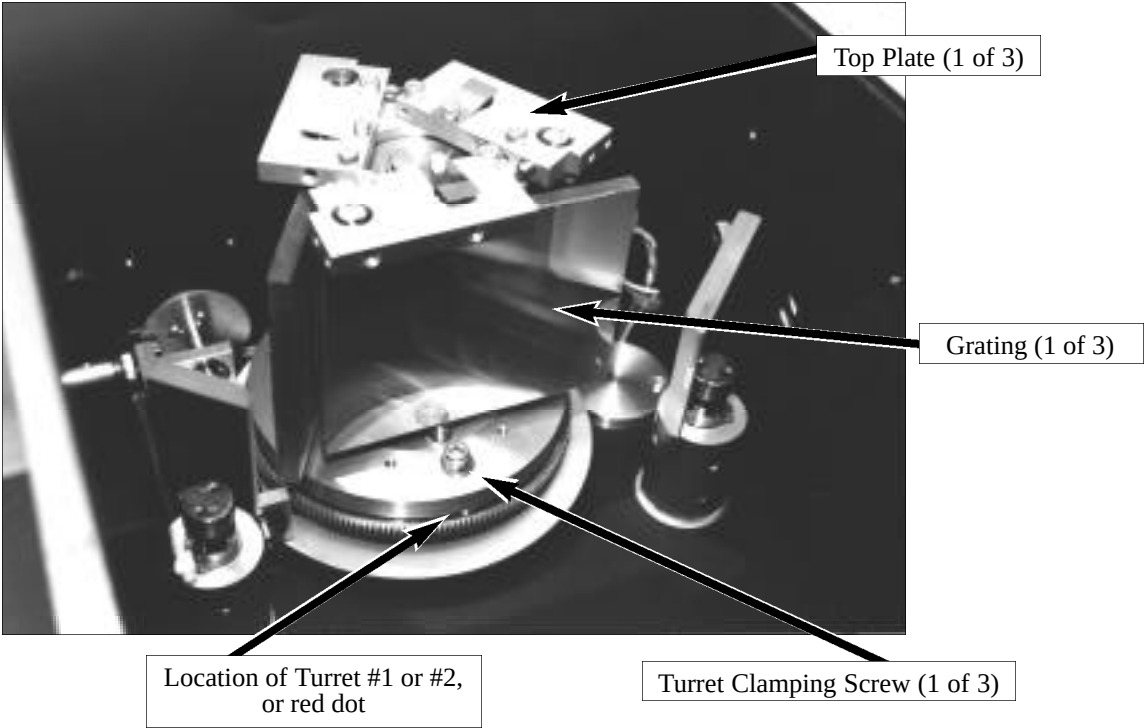


FIGURE 2

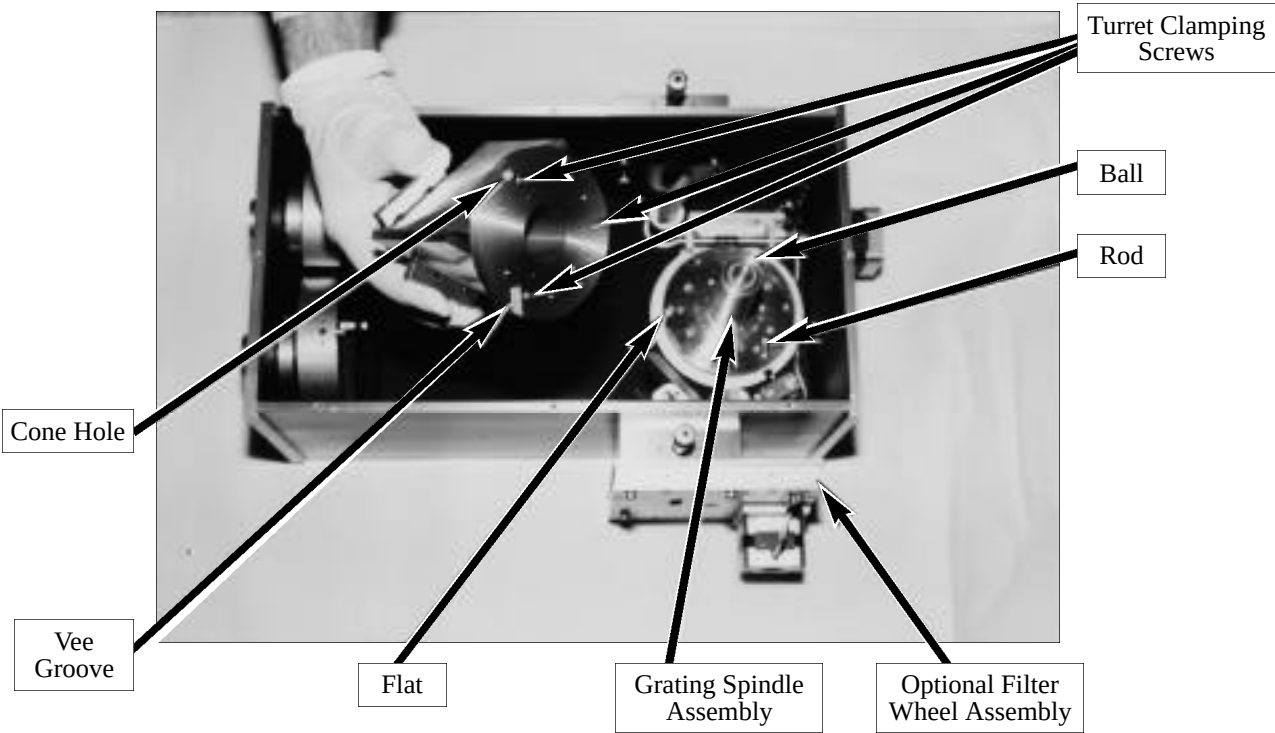


FIGURE 3

